Tutorial Week 4, STA2020F 2024: Non-parametric tests

1 Question 1

An article reports histamine levels in the saliva of two groups of people, those suffering from allergies and those without. The data below form part of this article.

With allergies	67.6	39.6	111.2		
Without	34.3	39.6	48.0	4.7	45.5

There is reason to doubt that these samples were drawn from normally distributed populations.

Carry out a suitable test of hypothesis to decide if the two groups differ in histamine levels. Be sure to state null and alternative hypotheses, show how you calculate the test statistic and provide a clear conclusion to the test, including a p-value.

2 Question 2

Property taxes are based on assessed values of property. In South Africa, the law requires that the values be "at or near" the market value of the property. In Rondebosch, a tax protest group has claimed that assessed values are higher than market values. To address this claim, the local tax assessor, together with representatives from the protest group, has selected 10 properties at random in the Rondebosch area that have sold within the past 6 months. Both parties agree that the sales price was the market value at the time of sale. The data (in thousands of rands) are presented in the table below.

Property	Assessed Value	Market Value
1	1560	1495
2	2200	2350
3	965	1250
4	1360	1275
5	1600	1750
6	1950	1800
7	3450	3650
8	1680	1655
9	4450	4150
10	2550	2700

Is there evidence to support the protest group's claim? Test at the 5% significance level. Make sure to include the following in your solution:

1. Identifying and giving reasons for the most suitable non-parametric test
2. Stating the correct hypotheses
3. Computing the test statistic
4. Determining the critical region
5. Stating the correct conclusion

3 Question 3

The following are the final examination grades of samples from three groups of students who where taught Swahili by one of three different methods (classroom instruction and language laboratory (A), only classroom instruction (B), and only self-study in language laboratory (C)):

A	94	88	91	74	87	97	
B	85	82	79	84	61	72	80
C	89	67	72	76	69		

Are the three methods are equally effective? Identify (with reasons) and conduct a suitable test. Use a 5% significance level, and make use of the p-value approach when making your conclusion.